

Element C:

This list is in order of priority.

High Priority

1. Our lock must keep the door within 1/4" of the stall wall.

Explanation: This requirement ties to prior research because the problem at hand is that the child has a fear of others watching them use the restroom. If the lock does not work they will not be able to use the restroom during the trip. This distance is compliant with the Americans with Disabilities Act (ADA) mandates that stalls have a 1/4" to 1" gap so that people's fingers do not get stuck.

2. Our lock must be able to withstand 50 Newtons of force.

Explanation: The stakeholder will need to use our device multiple times over their trip, and it is vital that it does not break when it is deployed/removed.

3. Our lock must be reusable for at least 20 uses.

Explanation: The child uses the restroom approximately 5 times a day. This is a weekend trip so we approximate the child will use the restroom under 20 times (Source 2).

4. Our lock must hold the door shut for 5 minutes at a time

Explanation: the lock needs to do this so that the child feels confident in their ability to pee without being exposed. If they do not feel confident, their paruresis will worsen and their self confidence will drop.

5. Our lock must weigh less than one pound and meet all TSA requirements.

Explanation: The stakeholders are traveling via plane so the product must be able to get past TSA. The stakeholder is also going to the mountains and an amusement park, two places where walking is common, so the product needs to be under one pound to not be burdensome on its carrier.

Low priority

6. Our lock must be made smooth so that it can be sanitized with a wipe.

The lock must be comfortable to wipe, above a 7 on a scale of 1-10 when surveyed.

Explanation: Bathroom surfaces harbor a lot of bacteria, so it is vital that our apparatus has a smooth surface for quick cleaning. Moreover, harmful bacteria could enter the stakeholder's purse and cause them to get sick if they cannot clean it with the sweep of a wipe.

7. Our lock must be able to be attached and removed in 15 seconds.

Explanation: The lock must be deployed and detached in less than 15 seconds so that other people in line do not get annoyed, and so that the child does not feel embarrassed when using the restroom.

8. The lock must not exceed \$25 to purchase

Explanation: Home Depot sells a good quality portable lock for 23 dollars, so we can assume that our lock being below 25 dollars will allow it to thrive in the market.

Works Cited:

Source 1: Kuske, Morgan. "ADA Bathroom Requirements: Restroom Space and Toilet Compartments." *LaForce, LLC*, 13 Jan. 2021, laforceinc.com/blog/ada-bathroom-requirements-restroom-space-and-toilet-compartments/.

Source 2: Greene, Lori. "Door Opening Force." *I Dig Hardware*, 24 Feb. 2009, idighardware.com/2009/02/door-opening-force/.

Source 3: Interview with the stakeholder

Source 4: Interview with stakeholder, "Shy Bladder Syndrome Can Be Debilitating." *Uclahealth.org*, UCLA Health, 13 June 2022, www.uclahealth.org/news/article/shy-bladder-syndrome-can-be-debilitating.

Source 5: Interview with stakeholder, Transportation Security Administration. "What Can I Bring?" *Transportation Security Administration*, 2019, www.tsa.gov/travel/security-screening/whatcanibring/all.

Source 6: University of Florida. "EML2322L -- Friction Coefficients." *Ufl.edu*, 2024, web.mae.ufl.edu/designlab/Class%20Projects/Background%20Information/Friction%20coefficients.htm.

Source 7: “Platonic Work Relationship Question Bathroom Closest My Desk Is Single Stall with Occupied Dial Lock. I Saw the Time before Use Say Noon While Still Clocked Out. I Used the Bathroom the Entire Time I Was in There. Heard a Voice Loudly on the Other Side Say “Well, I Guess There Is” and a Very Loud Banging on the Door After. I Was Drying My Hands at the Time of This Noise. I Came out and Checked the Time When I Got back to My Desk 12:05. I Think This Was Rude and Unneeded. Thoughts? Feedback?” *Glassdoor*, 9 May 2024, www.glassdoor.com.au/Community/relationships/platonic-work-relationship-questionbathroom-closest-my-desk-is-single-stall-with-occupied-dial-lock-i-saw-the-time-before. Accessed 16 Jan. 2026.

Source 8: “WINSOON Portable Door Lock.” *Homedepot.com*, 2026, <https://www.homedepot.com/p/WINSOON-Portable-Door-Lock-Black-Home-Security-Door-Lock-Travel-Lockdown-Locks-for-Additional-Safety-GCM6043/330894523?source=shoppingads&locale=en-US&fp=ggl>